SafePay MENA

The AI Fraud Shield for Instant Payments

Theme 4: Secure Fintech, Payments & Anti-Fraud Innovation | GSMA Open Gateway

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"1.1 billion instant transactions in Egypt.

Every single one is irreversible.

What happens when the wrong person hits send?"

The MENA Payment Boom Has a Fraud Problem

$275.47B	1.1B	12.5M	$469.9M	64%
MENA Digital Payments 2026	Egypt InstaPay Tx H1'25	UAE Aani Users	Saudi Fraud Spend '25	Arab Adults Unbanked

- **Instant payments = instant fraud risk:** Egypt's InstaPay and UAE's Aani process non-refundable push transfers.
- **Trust is the barrier:** 200M Arab adults remain unbanked. Fraud undermines trust in digital finance.
- **Massive investment:** Saudi Arabia alone spent $470M on fraud detection infrastructure in 2025.

How SIM Swap Fraud Destroys Accounts

Step 1: Social Engineer	Step 2: SIM Swap	Step 3: Intercept OTP	Step 4: Drain Account
Fraudster calls carrier: 'I lost my SIM card'	Carrier transfers victim's number to fraudster's SIM	Fraudster receives bank's SMS 2FA codes	Instant transfer drains account. **Irreversible.**

- **Individual losses:** Range from $270 to $160,000+ per SIM swap incident.
- **UAE Case:** $1.5 million stolen in a single attack combining SIM swap + insider collusion.
- **INTERPOL Action:** Operation Ramz (2025-2026) arrested 200+ suspects across 13 MENA countries.

The Fraud Gap: Banks & Telecoms Don't Talk

BANK WORLD	THE GAP	TELECOM WORLD
• Sees transaction amount, recipient, timestamp **BLIND TO:** SIM swaps, device changes, roaming status, network-level authentication	❌ No real-time data sharing framework between banks and carriers	• Sees SIM swaps, IMEI changes, roaming status, cell tower locations **BLIND TO:** Bank transfers and payment authorization flows

PwC 2025 GCC Fraud Report: "The critical intelligence gap between telecoms and financial institutions is the #1 vulnerability in digital banking."

SafePay MENA: The Bridge

An AI agent that asks the telecom network **'Is this person legit?'** before every high-risk payment.

Detect	Decide	Defend
Queries 4 CAMARA APIs via Nokia Network-as-Code for real-time carrier intelligence.	AI Agent (Google ADK + Gemini Flash) weighs signals into a risk score (0-100).	Autonomous decision: APPROVE (silent), STEP-UP, or BLOCK - before money moves.

- **Zero friction:** Low-risk payments proceed silently without user interaction.
- **Carrier-level security:** Replaces interceptable SMS OTPs with silent network authentication.
- **Audit trail:** Generates natural-language reasoning for regulatory compliance.

4 CAMARA APIs. One Intelligent Decision.

API	Endpoint	Fraud Signal Checked	Weight
SIM Swap	POST /sim-swap/v0/check	Was SIM card swapped recently? Swapped <24h = RED FLAG	**35%**
Number Verification	POST /number-verification/v0/verify	Silent carrier auth: does device match phone number? Replaces SMS OTP.	**30%**
Device Status	POST /device-status/v0/roaming	Is device reachable? Is it roaming in an unexpected country?	**15%**
Device Swap	POST /device-swap/v0/check	Was SIM moved to a new physical handset (different IMEI)?	**20%**

All APIs accessed via **Nokia Network-as-Code** platform using official Python SDK.

Not a Pipeline. An Agent.

Rule-Based Pipeline ■	SafePay AI Agent ■
• Calls ALL APIs on every transaction • Fixed if/else decision tree • Returns generic pass/fail • Same behavior for $10 and $50,000 • Cannot handle edge cases or context	• **Plans:** Decides WHICH APIs to call based on risk context • **Reasons:** Weighs multiple signals dynamically • **Explains:** Generates natural-language reasoning trace • **Adapts:** Low-risk = 1 check; High-risk = 4 checks • **Approved tooling:** Built with Google ADK + Gemini 2.0 Flash

Three Scenarios. Three Decisions.

Scenario A: Normal	Scenario B: Attack	Scenario C: Ambiguous
Tx: 500 EGP to Mom (2 PM) SIM Swap: No (0.0) Num Verify: Match (0.0) Context: Known recipient (0.05) **Score: 1 / 100** ■ **APPROVE (silent)**	**Tx:** 45,000 EGP to stranger (4 AM) SIM Swap: 3h ago (1.0) Num Verify: FAILED (1.0) Roaming: Nigeria (0.7) **Score: 93 / 100** ■ **BLOCK + ALERT**	**Tx:** 8,000 EGP to new contact (11 AM) SIM Swap: 5 days ago (0.5) Num Verify: Match (0.0) Context: New recipient (0.3) **Score: 24 / 100** ■ **APPROVE (legit swap)**

Technical Architecture

Payment Platform	SafePay Backend + Agent	Nokia NaC Sandbox
InstaPay / Aani / stc pay	**FastAPI + Google ADK + Gemini Flash**	4 CAMARA APIs
Sends transaction JSON	Orchestrates APIs & Risk Scoring Engine	Provides network signals

Supabase (PostgreSQL)	Next.js Frontend Dashboard
Audit Log table (decision traces) + User Profiles table	Real-time risk gauge + Agent reasoning trace viewer

Market Opportunity

TAM	SAM	SOM
$275.47B	**$6.35B**	**$44M**
MENA Digital Payments Market (2026)	MENA Fintech Market Size (2026)	Egypt InstaPay alone at $0.02/check

- **22 MENA Markets:** Every Arab League country + Turkiye has expanding digital payment systems.
- **80+ Operators:** Globally committed to Open Gateway, making carrier-agnostic deployment seamless.
- **Growth Runway:** Reducing fraud unlocks financial inclusion for 200 million unbanked Arabs.

Business **Model**

Per-Transaction Fee	Monthly Subscription	Enterprise License
$0.01 - $0.05	**$500 - $5,000**	**$50K - $200K**
Per fraud check. Paid by banks & fintechs.	Monthly tiered by volume. Mid-size banks & wallets.	Annual license. Central banks & large banks.

Unit Economics: Cost of fraud = $270 - $160,000+ per incident. Cost of SafePay check = <$0.01 on Nokia NaC. SafePay pays for itself if it prevents just **1 fraud case per 10,000 transactions**.

Impact & Scalability

Quantified Impact (Pilot Data)	Scalability Blueprint
• **-44% Scam Losses:** Proven in early CAMARA global pilot deployments. • **-55% False Positives:** Fewer legitimate transfers blocked unnecessarily. • **<$0.01 Cost per Check:** Highly economical compared to $15-25 manual investigations. • **$270-$160K Saved:** Per prevented SIM swap attack.	• **Carrier-Agnostic:** CAMARA standards mean 1 integration works across e&, stc, Ooredoo, Vodafone. • **Payment-Agnostic:** Integrates with InstaPay, Aani, stc pay, Fawry. • **200M Unbanked:** Fraud prevention builds the trust needed for digital onboarding.

Why SafePay Wins

What Exists Today	The Unsolved Missing Link	SafePay MENA
Individual CAMARA API launches (e& UAE, Ooredoo Qatar)	No intelligent orchestration layer	**Multi-API AI Agent** that weighs signals together
Bank transaction monitoring systems	Zero telecom network signals	**Bridges the fraud gap** between banks & carriers
SMS OTP for authentication	Interceptable, weakest link	**Carrier Number Verification** (silent & unforgeable)
GSMA + FICO Scam Signal specification	Specification stage, not deployed	**Working prototype** with 4 integrated APIs

"The telecom APIs exist individually. SafePay is the intelligent brain that orchestrates them."

The technology exists.
The market is screaming for it.

SafePay is the bridge between $275 billion in payments and the network that can protect them.

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